

Zeta-Law Entropy, Gamma Curvature, and Tail Partition Applications

Pudim AI Project

Abstract

We study the Riemann zeta function as the partition function of the probability law $\rho_\beta(n) = n^{-\beta}/\zeta(\beta)$ on the positive integers. This normalization turns zeta ratios into moments, logarithmic derivatives into energy cumulants, divisor identities into probabilistic decompositions, and zeta tails into reciprocal partition problems. The first main result identifies the microscopic successor entropy of the zeta law with the limit and supremum of finite modular successor entropies; for prime moduli these modular shadows are explicit nonlinear functionals of Dirichlet L -values. The second structural result is an Euler-score decomposition expressing $-(\log \zeta)'(\beta)$ as a von Mangoldt-weighted divisibility average. We then prove nine applications: the Alzer–Kwong convexity and concavity pattern for $1/\zeta$ on negative real intervals; positivity of a Nantomah expression involving $\zeta(n+1), \zeta(n+2), \zeta(n+3)$; a generalized Holder inequality for $\Gamma(s)\zeta(s)$ extending Sroysang’s formulation; complete monotonicity of $(\log \Gamma(x) + \log \Gamma(1/x))''$; a sharp threshold for reciprocal Gamma-product monotonicity; complete monotonicity of $-(\psi(x)\psi(1/x))''$; a counterexample showing that the higher-polygamma strengthening P_n'' completely monotone for all n fails; and exact inverse-tail floor formulas for $\zeta_n(7)^{-1}$ and $\zeta_n(8)^{-1}$, where $\zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}$. The framework gives a compact route from zeta probability laws to entropy, curvature, Mellin-integral inequalities, reciprocal Gamma products, and discrete reciprocal partition formulas.

Keywords. Riemann zeta function; zeta probability law; Gibbs measure; entropy; modular residues; Dirichlet L -functions; von Mangoldt function; reciprocal zeta; zeta tails; Gamma function; digamma function; polygamma functions; complete monotonicity; Holder inequality.

1 Motivation and main results

Many classical identities for $\zeta(s)$ become more transparent after a probabilistic normalization. For $\beta > 1$, the weights $n^{-\beta}$ define a Gibbs law whose partition function is precisely $\zeta(\beta)$. The energy is $\log n$, the free energy is $\log \zeta(\beta)$, and ratios such as $\zeta(\beta+r)/\zeta(\beta)$ become moments. This viewpoint is elementary, but it packages several unrelated-looking zeta inequalities into a common calculus.

The paper develops five connected layers. The moment layer records basic zeta-law identities. The modular entropy layer compares the successor map $n \mapsto n+1$ with its finite residue shadows modulo q . The Euler-score layer rewrites zeta energy through von Mangoldt divisibility events. The Mellin-Planck layer treats $\Gamma(s)\zeta(s)$ as a continuous partition function and converts generalized Holder inequalities into integral norm inequalities. The tail layer studies $\zeta_n(s)^{-1}$ as a reciprocal partition problem for the truncated zeta law.

Section 2 records the nine solved external problems, partial resolutions, or candidate-problem specializations with stable application labels. APP-0001 is the Alzer–Kwong reciprocal-zeta conjecture, proved by transporting a positive-axis curvature estimate through the functional equation. APP-0002 is the Nantomah positivity question, proved by splitting a zeta expression into a positive linear tail and a controlled quadratic correction. APP-0003 is the Sroysang generalized Holder

problem, proved directly from generalized Holder applied to the Planck kernel. APP-0004 is a reciprocal-Gamma complete-monotonicity problem, solved by a Weierstrass-product Laplace kernel. APP-0007 is the reciprocal-digamma product problem of Qi–Lim–Nantomah, solved by a grouped Laplace kernel for $-(\psi(x)\psi(1/x))''$. APP-0008 refutes the stronger complete-monotonicity part of the higher-polygamma product problem by proving $P_n'''(2) > 0$ for all $n \geq 29$, with an additional lower-order certificate $P_7^{(6)}(3) < 0$. APP-0005 and APP-0006 give exact formulas for $\lfloor \zeta_n(7)^{-1} \rfloor$ and $\lfloor \zeta_n(8)^{-1} \rfloor$, resolving the first two local tail-inversion special cases while leaving the general integer $s > 6$ problem open. APP-0009 closes the endpoint and necessity gap in Bulboaca–Zayed’s reciprocal Gamma-product threshold problem by proving the exact condition $\psi(1 + \rho) \geq \gamma$.

2 Applications

The following application labels are used consistently in the repository landing page, the application changelog, and the theorem names below. The order here is the landing-page order; the proof bodies are placed where their methods naturally fit the paper.

1. **APP-0001.** Alzer–Kwong convexity and concavity problem. Solved by Theorem 8.2; source problem reference [3].
2. **APP-0002.** Nantomah zeta positivity problem. Solved by Theorem 8.1; source problem reference [1].
3. **APP-0003.** Sroysang generalized Holder problem. Solved by Theorem 8.3; source problem reference [2].
4. **APP-0004.** Complete monotonicity of $(\log \Gamma(x) + \log \Gamma(1/x))''$. Solved by Theorem 7.5; source problem reference [4].
5. **APP-0005.** Exact inverse-tail floor formula at $s = 7$. Solved by Theorem 4.2; source context [6, 7].
6. **APP-0006.** Exact inverse-tail floor formula at $s = 8$. Solved by Theorem 4.3; source context [6, 7].
7. **APP-0007.** Complete monotonicity of $-(\psi(x)\psi(1/x))''$. Solved by Theorem 3.4; source problem reference [4].
8. **APP-0008.** Counterexample to complete monotonicity of higher-order polygamma product curvature. Solved by Theorem 3.5; source problem reference [4].
9. **APP-0009.** Sharp reciprocal Gamma-product monotonicity threshold. Solved by Theorem 7.4; source problem reference [5].

3 Notation and zeta-law calculus

Definition 3.1 (Riemann zeta probability law). For $\beta > 1$, define a probability law ρ_β on $\mathbb{N}$ by

$$\rho_\beta(n) = \frac{n^{-\beta}}{\zeta(\beta)}, \quad n \in \mathbb{N}. \quad (3.1)$$

Equivalently, with $E(n) = \log n$ and $Z(\beta) = \zeta(\beta)$,

$$\rho_\beta(n) = \frac{e^{-\beta E(n)}}{Z(\beta)}. \quad (3.2)$$

Definition 3.2 (Zeta free energy). The zeta free energy is

$$A(\beta) = \log \zeta(\beta), \quad \beta > 1. \quad (3.3)$$

Proposition 3.3 (Free-energy derivatives). For every $\beta > 1$,

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \quad A''(\beta) = \text{Var}_\beta(\log N). \quad (3.4)$$

Proof. Absolute convergence of the differentiated Dirichlet series on compact subintervals of $(1, \infty)$ permits termwise differentiation. Thus

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\frac{\sum_{n \geq 1} (\log n) n^{-\beta}}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N]. \quad (3.5)$$

Differentiating once more gives

$$A''(\beta) = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2 = \text{Var}_\beta(\log N). \quad (3.6)$$

□

The same source asks whether $P_0(x) = \psi(x)\psi(1/x)$ is concave, and more strongly whether $-P_0''$ is completely monotonic on $(0, \infty)$. The next theorem answers the stronger question.

Theorem 3.4 (APP-0007: Complete monotonicity of reciprocal digamma product curvature). Let

$$P_0(x) = \psi(x)\psi(1/x), \quad x > 0. \quad (3.7)$$

Then $-P_0''$ is strictly completely monotonic on $(0, \infty)$. In particular, P_0 is strictly concave on $(0, \infty)$.

Proof. Direct differentiation gives

$$\begin{aligned} P_0''(x) &= \psi''(x)\psi(1/x) - \frac{2}{x^2}\psi'(x)\psi'(1/x) \\ &\quad + \frac{2}{x^3}\psi(x)\psi'(1/x) + \frac{1}{x^4}\psi(x)\psi''(1/x). \end{aligned} \quad (3.8)$$

Use the shifted Weierstrass expansion

$$\psi(x) = -\gamma - \frac{1}{x} + \sum_{m=1}^{\infty} a_m(x), \quad a_m(x) = \frac{x}{m(m+x)}, \quad (3.9)$$

and, after substituting $1/x$,

$$\psi(1/x) = -\gamma - x + \sum_{n=1}^{\infty} b_n(x), \quad b_n(x) = \frac{1}{n(nx+1)}. \quad (3.10)$$

On compact subintervals of $(0, \infty)$, the series for a_m , b_n , and their first two derivatives are uniformly summable; the differentiated double products are dominated by products of summable majorants. Hence P_0 may be differentiated term by term twice.

Let $A_0(x) = -\gamma - 1/x$ and $B_0(x) = -\gamma - x$. The base term satisfies

$$-\frac{d^2}{dx^2}(A_0(x)B_0(x)) = -\frac{2\gamma}{x^3}. \quad (3.11)$$

For $n \geq 1$, elementary rational differentiation and Laplace inversion give

$$-\frac{d^2}{dx^2}(A_0(x)b_n(x)) = \int_0^\infty e^{-xt} t^2 \frac{n + (\gamma - n)e^{-t/n}}{n^2} dt, \quad (3.12)$$

and, for $m \geq 1$,

$$-\frac{d^2}{dx^2}(B_0(x)a_m(x)) = \int_0^\infty e^{-xt} t^2 (m - \gamma) e^{-mt} dt. \quad (3.13)$$

For $mn > 1$,

$$-\frac{d^2}{dx^2}(a_m(x)b_n(x)) = \int_0^\infty e^{-xt} t^2 \left[\frac{e^{-t/n}}{mn^2(mn - 1)} - \frac{e^{-mt}}{n(mn - 1)} \right] dt. \quad (3.14)$$

The exceptional product is $a_1(x)b_1(x) = x/(x+1)^2$, and

$$-\frac{d^2}{dx^2} \frac{x}{(x+1)^2} = \int_0^\infty e^{-xt} t^2 (t-1) e^{-t} dt. \quad (3.15)$$

The identities above follow from

$$\frac{1}{(x+c)^3} = \int_0^\infty e^{-xt} \frac{t^2}{2} e^{-ct} dt, \quad \frac{1}{(x+c)^4} = \int_0^\infty e^{-xt} \frac{t^3}{6} e^{-ct} dt. \quad (3.16)$$

The coefficients of the double products group by the two identities

$$\sum_{m=1}^\infty \frac{1}{m(mn-1)} = -\psi(1-1/n) - \gamma \quad (n \geq 2) \quad (3.17)$$

and

$$\sum_{n=1}^\infty \frac{1}{n(mn-1)} = -\psi(1-1/m) - \gamma \quad (m \geq 2). \quad (3.18)$$

Both are immediate from $\psi(z+1) + \gamma = \sum_{k=1}^\infty z/(k(k+z))$, with $z = -1/n$ and $z = -1/m$. Combining (3.11)–(3.18) gives

$$-P_0''(x) = \int_0^\infty e^{-xt} K(t) dt, \quad (3.19)$$

where

$$K(t) = t^2 \left[1 - \gamma + (t-1)e^{-t} + \sum_{n=2}^\infty \frac{n(1-e^{-t/n}) - \psi(1-1/n)e^{-t/n}}{n^2} + \sum_{m=2}^\infty (m + \psi(1-1/m))e^{-mt} \right]. \quad (3.20)$$

It remains to prove $K(t) > 0$. Since ψ is increasing and $1-1/n \in [1/2, 1)$,

$$-\psi(1-1/n) > 0 \quad (n \geq 2). \quad (3.21)$$

Also

$$m + \psi(1 - 1/m) \geq 2 - \gamma - 2 \log 2 > 0 \quad (m \geq 2), \quad (3.22)$$

because $\psi(1 - 1/m) \geq \psi(1/2) = -\gamma - 2 \log 2$. Thus the two sums in (3.20) are positive term by term, apart from the possible need to dominate $1 - \gamma + (t - 1)e^{-t}$.

For $n \geq 2$,

$$-\psi(1 - 1/n) = \gamma + \int_{1-1/n}^1 \psi'(u) du > \gamma + \int_{1-1/n}^1 \frac{du}{u^2} = \gamma + \frac{1}{n-1}. \quad (3.23)$$

Consequently the n -sum in (3.20) is bounded below by

$$Ae^{-t/2}, \quad A = \gamma(\zeta(2) - 1) + 2 - \zeta(2). \quad (3.24)$$

Therefore $K(t)/t^2$ is bounded below by

$$f(t) = 1 - \gamma + (t - 1)e^{-t} + Ae^{-t/2}. \quad (3.25)$$

Now

$$f(0) = A - \gamma = (2 - \zeta(2))(1 - \gamma) > 0, \quad \lim_{t \rightarrow \infty} f(t) = 1 - \gamma > 0. \quad (3.26)$$

Moreover,

$$f'(t) = e^{-t} \left(2 - t - \frac{A}{2} e^{t/2} \right), \quad (3.27)$$

and the factor in parentheses is strictly decreasing. Hence f has at most one critical point, and any such point is a maximum. Its minimum on $[0, \infty)$ is attained at an endpoint, so $f(t) > 0$. Thus $K(t) > 0$ for $t > 0$.

Differentiating under (3.19) yields

$$(-1)^r \frac{d^r}{dx^r} (-P_0''(x)) = \int_0^\infty t^r e^{-xt} K(t) dt > 0 \quad (3.28)$$

for all $r \geq 0$. This proves strict complete monotonicity of $-P_0''$, and the case $r = 0$ gives strict concavity of P_0 . $\square$

Qi, Lim, and Nantomah also ask whether the higher-order products $P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x)$, $n \in \mathbb{N}$, are convex, and more strongly whether P_n'' is completely monotonic. The following result refutes the stronger assertion. It does not settle the weaker convexity question.

Theorem 3.5 (APP-0008: Counterexample to complete monotonicity of higher-order polygamma product curvature). *Let*

$$P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x), \quad n \in \mathbb{N}, \quad x > 0. \quad (3.29)$$

Then P_n'' is not completely monotonic for all $n \in \mathbb{N}$. More precisely, for every $n \geq 29$,

$$P_n'''(2) > 0, \quad (3.30)$$

and, already at lower order,

$$P_7^{(6)}(3) < 0. \quad (3.31)$$

Proof. For $m \geq 1$, define the sign-normalized polygamma function

$$A_m(x) = (-1)^{m+1} \psi^{(m)}(x) = m! \sum_{k=0}^{\infty} (x+k)^{-m-1}. \quad (3.32)$$

Then $A_m(x) > 0$, $A'_m(x) = -A_{m+1}(x)$, and $P_n(x) = A_n(x)A_n(1/x)$. Differentiating twice gives

$$\begin{aligned} P_n''(x) &= A_{n+2}(x)A_n(1/x) - \frac{2}{x^2}A_{n+1}(x)A_{n+1}(1/x) \\ &\quad - \frac{2}{x^3}A_n(x)A_{n+1}(1/x) + \frac{1}{x^4}A_n(x)A_{n+2}(1/x). \end{aligned} \quad (3.33)$$

First prove (3.30). Put $p = n + 1$, and normalize

$$Q_p(x) = \frac{P_n(x)}{(n!)^2} = \sum_{m, \ell \geq 0} \left(\frac{x}{(x+m)(1+\ell x)} \right)^p. \quad (3.34)$$

At $x = 2$, the term $(m, \ell) = (0, 0)$ is constant, and the unique largest nonconstant base is $(m, \ell) = (1, 0)$, equal to $2/3$. Write

$$Q_p(x) = 1 + \left(\frac{x}{x+1} \right)^p + R_p(x). \quad (3.35)$$

For $h(x) = x/(x+1)$, direct differentiation gives

$$\frac{d^3}{dx^3} h(x)^p = p \left(\frac{x}{x+1} \right)^p \frac{p^2 - 6px - 3p + 6x^2 + 6x + 2}{x^3(x+1)^3}. \quad (3.36)$$

Therefore

$$\left. \frac{d^3}{dx^3} h(x)^p \right|_{x=2} = \frac{p(p^2 - 15p + 38)}{216} \left(\frac{2}{3} \right)^p. \quad (3.37)$$

For $p \geq 30$, this is at least

$$\frac{p^3}{432} \left(\frac{2}{3} \right)^p. \quad (3.38)$$

For a general summand

$$F_{m, \ell}(x) = \left(\frac{x}{(x+m)(1+\ell x)} \right)^p, \quad (3.39)$$

let $u_{m, \ell} = d(\log F_{m, \ell}^{1/p})/dx$. At $x = 2$,

$$|u_{m, \ell}| \leq \frac{1}{2}, \quad |u'_{m, \ell}| \leq \frac{1}{4}, \quad |u''_{m, \ell}| \leq \frac{1}{4}. \quad (3.40)$$

Since

$$F_{m, \ell}''' = F_{m, \ell} (p^3 u_{m, \ell}^3 + 3p^2 u_{m, \ell} u'_{m, \ell} + p u''_{m, \ell}), \quad (3.41)$$

we have $|F_{m, \ell}'''(2)| \leq p^3 F_{m, \ell}(2)$ for $p \geq 2$. The remaining mass in R_p is bounded by

$$\sum_{(m, \ell) \neq (0, 0), (1, 0)} F_{m, \ell}(2) < 8 \cdot 2^{-p} \quad (p \geq 3). \quad (3.42)$$

Indeed, the $m \geq 2, \ell = 0$ part is at most $3 \cdot 2^{-p}$, while the $\ell \geq 1$ part is at most $2 \cdot 2^{-p}(1 + (2/3)^p + 3 \cdot 2^{-p})$. Combining (3.38) and (3.42),

$$Q_p'''(2) \geq p^3 2^{-p} \left[\frac{1}{432} \left(\frac{4}{3} \right)^p - 8 \right]. \quad (3.43)$$

Because $(4/3)^{30} > 432 \cdot 8$, (3.43) is positive for $p \geq 30$. Hence $P_n'''(2) = (n!)^2 Q_{n+1}'''(2) > 0$ for $n \geq 29$. Complete monotonicity of P_n'' would require $(P_n'')' = P_n''' \leq 0$, so P_n'' is not completely monotonic for $n \geq 29$.

We now record the lower-order obstruction (3.31). For bookkeeping, put

$$T_{p,i,j}(x) = x^{-p} A_i(x) A_j(1/x). \quad (3.44)$$

The identity $A'_m = -A_{m+1}$ gives the exact recurrence

$$\frac{d}{dx} T_{p,i,j}(x) = -p T_{p+1,i,j}(x) - T_{p,i+1,j}(x) + T_{p+2,i,j+1}(x). \quad (3.45)$$

Starting with (3.33), applying (3.45) four times with $n = 7$ expresses $P_7^{(6)}(3)$ as an exact finite linear combination of 22 terms $T_{p,i,j}(3)$.

The sign of this finite expression can be certified by rational intervals. For rational $x > 0$ and $N \geq 1$, monotone integral comparison in (3.32) gives

$$m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} \right) \leq A_m(x), \quad (3.46)$$

and

$$A_m(x) \leq m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} + (x+N)^{-m-1} \right). \quad (3.47)$$

Using $N = 80$ in (3.46) and (3.47) for every $A_m(3)$ and $A_m(1/3)$ appearing in the 22-term expression, and choosing upper or lower endpoints according to the sign of each coefficient, yields

$$-74998.154649422399828421756528409981771664439555587 < P_7^{(6)}(3), \quad (3.48)$$

and

$$P_7^{(6)}(3) < -74997.331312439140215056900814096498575289999496021. \quad (3.49)$$

Thus $P_7^{(6)}(3) < 0$. If P_7'' were completely monotonic, then

$$(-1)^4 \frac{d^4}{dx^4} P_7''(3) = P_7^{(6)}(3) \geq 0, \quad (3.50)$$

contradicting (3.49). Hence the proposed complete-monotonicity strengthening for all $n \in \mathbb{N}$ is false in both an infinite high-order family and already at $n = 7$. $\square$

Proposition 3.6 (Zeta-law calculus). *Let $\beta > 1$, and let N have law ρ_β . For every $r \geq 0$,*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta+r)}{\zeta(\beta)}. \quad (3.51)$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha). \quad (3.52)$$

Proof. The moment identity follows by direct summation:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} n^{-r} \frac{n^{-\beta}}{\zeta(\beta)} = \frac{\zeta(\beta+r)}{\zeta(\beta)}. \quad (3.53)$$

For relative entropy, substitute (3.1):

$$D(\rho_\alpha \| \rho_\beta) = \sum_{n=1}^{\infty} \rho_\alpha(n) ((\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)), \quad (3.54)$$

which is (3.52). $\square$

4 Tail zeta laws and inverse-tail floors

Definition 4.1 (Tail zeta partition function). For $s > 1$ and $n \in \mathbb{N}$, define the tail zeta partition function by

$$Z_{s,n} = \zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}. \quad (4.1)$$

The corresponding tail zeta law is the probability distribution on $\{n, n+1, \dots\}$ given by

$$\rho_{s,n}(k) = \frac{k^{-s}}{\zeta_n(s)}, \quad k \geq n. \quad (4.2)$$

Exact formulas for integer parts of reciprocal zeta tails have been studied for small integer exponents. Known formulas include $s = 2, 3, 4, 5$, with a sufficiently-large- n formula for $s = 6$; Kim and Song record that no analogous formula was known for integer $s > 6$ at the time of their work [6]. The 2024 asymptotic inverse-tail results of Pan and Wu provide related context [7]. The next two theorems give exact formulas at $s = 7$ and $s = 8$. The first uses a single rational telescoping enclosure, while the second needs adjacent corrected asymptotic truncations.

Theorem 4.2 (APP-0005: Exact inverse-tail floor formula at $s=7$). *Let*

$$T_7(n) = \zeta_n(7) = \sum_{k=n}^{\infty} \frac{1}{k^7}. \quad (4.3)$$

Define

$$Q(n) = 120n^6 - 360n^5 + 660n^4 - 720n^3 + 354n^2 - 54n + 375, \quad P(n) = \frac{Q(n)}{20}. \quad (4.4)$$

Then, for every $n \geq 28$,

$$\lfloor T_7(n)^{-1} \rfloor = \lfloor P(n) \rfloor. \quad (4.5)$$

For $1 \leq n \leq 27$, the exact values are

n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$
1	0	10	4495760	19	241765529
2	119	11	8167814	20	331399044
3	1862	12	14061542	21	447175152
4	12573	13	23143975	22	594869693
5	54069	14	36668777	23	781167220
6	175597	15	56228085	24	1013751716
7	471118	16	83808666	25	1301401638
8	1100904	17	121852400	26	1654089273
9	2317459	18	173321080	27	2083084422

(4.6)

Proof. The polynomial Q is positive on $\mathbb{N}$, because

$$Q(n) = 120(n-1)^6 + 360(n-1)^5 + 660(n-1)^4 + 720(n-1)^3 + 354(n-1)^2 + 54(n-1) + 375. \quad (4.7)$$

For $k \geq 1$, direct expansion gives

$$\frac{1}{k^7} - \left(\frac{20}{Q(k)} - \frac{20}{Q(k+1)} \right) = \frac{9(60284k^4 + 29176k^2 + 15625)}{k^7 Q(k) Q(k+1)} > 0. \quad (4.8)$$

Summing (4.8) over $k \geq n$ yields

$$T_7(n) > \frac{20}{Q(n)} = \frac{1}{P(n)}, \quad (4.9)$$

and therefore $T_7(n)^{-1} < P(n)$.

For $k \geq 28$, another expansion gives

$$\left(\frac{20}{Q(k) - 3} - \frac{20}{Q(k+1) - 3} \right) - \frac{1}{k^7} = \frac{36(20k^6 - 14961k^4 - 7235k^2 - 3844)}{k^7(Q(k) - 3)(Q(k+1) - 3)}. \quad (4.10)$$

The numerator in (4.10) is positive for $k \geq 28$. Indeed, writing $k = m + 28$, it becomes

$$20m^6 + 3360m^5 + 220239m^4 + 7105168m^3 + 114013021m^2 + 751143512m + 436261580, \quad (4.11)$$

which has positive coefficients for $m \geq 0$. Hence, for $n \geq 28$,

$$T_7(n) < \frac{20}{Q(n) - 3} = \frac{1}{P(n) - 3/20}. \quad (4.12)$$

Combining (4.9) and (4.12) gives

$$P(n) - \frac{3}{20} < T_7(n)^{-1} < P(n). \quad (4.13)$$

Modulo 20,

$$Q(n) \equiv 14n^2 - 14n + 15 \pmod{20}. \quad (4.14)$$

Since $n(n-1) \pmod{10} \in \{0, 2, 6\}$, this implies

$$Q(n) \pmod{20} \in \{3, 15, 19\}. \quad (4.15)$$

Thus the fractional part of $P(n)$ is one of $3/20, 15/20, 19/20$. If $a = \lfloor P(n) \rfloor$, then $P(n) - 3/20 \geq a$, and (4.13) gives

$$a < T_7(n)^{-1} < a + 1. \quad (4.16)$$

This proves (4.5) for $n \geq 28$.

For the finite range, $T_7(1) > 1$, so $0 < T_7(1)^{-1} < 1$. For $2 \leq n \leq 27$, let M_n denote the corresponding value in (4.6). With $N = 1500$, exact rational arithmetic verifies

$$\sum_{k=n}^N \frac{1}{k^7} > \frac{1}{M_n + 1} \quad (4.17)$$

and

$$\sum_{k=n}^N \frac{1}{k^7} + \frac{1}{6N^6} < \frac{1}{M_n}. \quad (4.18)$$

The monotone integral estimate

$$\sum_{k=N+1}^{\infty} \frac{1}{k^7} < \int_N^{\infty} x^{-7} dx = \frac{1}{6N^6} \quad (4.19)$$

then gives

$$\frac{1}{M_n + 1} < T_7(n) < \frac{1}{M_n}. \quad (4.20)$$

Therefore $M_n < T_7(n)^{-1} < M_n + 1$, proving (4.6). The finite rational certificate is stored with the source as a reproducible calculation. $\square$

Theorem 4.3 (APP-0006: Exact inverse-tail floor formula at s=8). *Let*

$$T_8(n) = \zeta_n(8) = \sum_{k=n}^{\infty} \frac{1}{k^8}. \quad (4.21)$$

Define

$$\begin{aligned} A_8(n) = & 7n^7 - \frac{49}{2}n^6 + \frac{637}{12}n^5 - \frac{1715}{24}n^4 + \frac{7399}{144}n^3 - \frac{1715}{96}n^2 + \frac{69685}{1728}n - \frac{65611}{3456} \\ & - \frac{23764069}{103680n} - \frac{23764069}{207360n^2} + \frac{2102131409}{1244160n^3} + \frac{2149659547}{829440n^4} \\ & - \frac{203290800013}{14929920n^5} - \frac{1146003910541}{29859840n^6} + \frac{21587525169497}{179159040n^7}. \end{aligned} \quad (4.22)$$

Then, for every $n \geq 6$,

$$\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor. \quad (4.23)$$

For $1 \leq n \leq 5$, the exact values are

$$\begin{array}{c|ccccc} n & 1 & 2 & 3 & 4 & 5 \\ \hline \lfloor T_8(n)^{-1} \rfloor & 0 & 245 & 5844 & 53503 & 291407 \end{array}. \quad (4.24)$$

Proof. Set

$$B_8(n) = A_8(n) + \frac{200335112892761}{358318080n^8}. \quad (4.25)$$

For $C = A_8$ or B_8 , write $R_C(k) = 1/C(k)$. Exact denominator clearing gives the two telescoping inequalities

$$R_{A_8}(k) - R_{A_8}(k+1) - \frac{1}{k^8} > 0 \quad (k \geq 3) \quad (4.26)$$

and

$$\frac{1}{k^8} - (R_{B_8}(k) - R_{B_8}(k+1)) > 0 \quad (k \geq 1). \quad (4.27)$$

In (4.26), the cleared numerator is a polynomial in $k-3$ with all coefficients positive; in (4.27), the cleared numerator is a polynomial in $k-1$ with all coefficients positive. The exact coefficient verification is recorded in the rational certificate accompanying the manuscript.

Summing (4.26) over $k \geq n$ gives

$$T_8(n) < R_{A_8}(n) = \frac{1}{A_8(n)} \quad (4.28)$$

for $n \geq 3$, and hence $A_8(n) < T_8(n)^{-1}$. Summing (4.27) gives

$$T_8(n) > R_{B_8}(n) = \frac{1}{B_8(n)}, \quad (4.29)$$

and hence $T_8(n)^{-1} < B_8(n)$. Therefore, for $n \geq 6$,

$$A_8(n) < T_8(n)^{-1} < B_8(n). \quad (4.30)$$

It remains to see that no integer lies between $A_8(n)$ and $B_8(n)$. Exact integer arithmetic verifies

$$\lfloor A_8(n) \rfloor = \lfloor B_8(n) \rfloor \quad (6 \leq n \leq 10^6). \quad (4.31)$$

For $n > 10^6$, let $P_8(n)$ be the polynomial part of $A_8(n)$, through the constant term. The numerator of $P_8(n)$ modulo 3456 runs through the odd residue classes, so $P_8(n)$ is at distance at least $1/3456$ from the next integer. The correction $A_8(n) - P_8(n)$ is negative and has absolute value $< 1/3456$, while

$$0 < B_8(n) - A_8(n) < \frac{1}{3456}. \quad (4.32)$$

Thus $B_8(n)$ is still below the next integer after $A_8(n)$. Combining this with (4.30) proves (4.23).

For $1 \leq n \leq 5$, (4.24) is certified by exact rational partial sums. With $N = 300$, the displayed value M_n satisfies, for $2 \leq n \leq 5$,

$$\sum_{k=n}^N \frac{1}{k^8} > \frac{1}{M_n + 1} \quad (4.33)$$

and

$$\sum_{k=n}^N \frac{1}{k^8} + \frac{1}{7N^7} < \frac{1}{M_n}. \quad (4.34)$$

Since

$$\sum_{k=N+1}^{\infty} \frac{1}{k^8} < \int_N^{\infty} x^{-8} dx = \frac{1}{7N^7}, \quad (4.35)$$

we get $M_n < T_8(n)^{-1} < M_n + 1$. For $n = 1$, $T_8(1) > 1$, so the floor is 0. $\square$

5 Modular successor entropy

Definition 5.1 (Modular residue distribution and successor entropy). For $q \geq 2$, define the residue distribution of ρ_β modulo q by

$$\mu_{q,\beta}(a) = \sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n), \quad a \in \mathbb{Z}/q\mathbb{Z}. \quad (5.1)$$

Writing $a + 1$ cyclically in $\mathbb{Z}/q\mathbb{Z}$, define

$$B_q(\beta) = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}. \quad (5.2)$$

Theorem 5.2 (Zeta-law successor entropy and modular resolution). For every $\beta > 1$, define

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (5.3)$$

Then

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta + k), \quad (5.4)$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \rightarrow \infty} B_q(\beta). \quad (5.5)$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1 + \frac{1}{n}\right)^\beta, \quad (5.6)$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log \left(1 + \frac{1}{n}\right). \quad (5.7)$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \quad (0 < x \leq 1) \quad (5.8)$$

with Abel limiting at $x = 1$, and summing against $n^{-\beta}$, gives (5.4).

For the modular identity, fix $q \geq 2$. The log-sum inequality gives, for each residue class a ,

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} \geq \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n \equiv a \pmod{q}} \rho_\beta(n+1)}. \quad (5.9)$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, it is $\mu_{q,\beta}(1) - \rho_\beta(1) \leq \mu_{q,\beta}(1)$. Summing (5.9) over a gives

$$\Sigma(\beta) \geq B_q(\beta) \quad (q \geq 2). \quad (5.10)$$

Conversely, fix $M \in \mathbb{N}$. If $q > M + 1$, then for $1 \leq a \leq M + 1$,

$$\mu_{q,\beta}(a) = \rho_\beta(a) + O_{\beta,M}(q^{-\beta}), \quad (5.11)$$

and $\mu_{q,\beta}(0) = q^{-\beta}$. All cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term, which satisfies

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_\beta(q^{-\beta} \log q). \quad (5.12)$$

Therefore

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \sum_{n=1}^M \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (5.13)$$

Letting $M \rightarrow \infty$ gives $\liminf_q B_q(\beta) \geq \Sigma(\beta)$, which together with (5.10) proves (5.5). $\square$

Corollary 5.3 (Prime-modulus Dirichlet L-resolution). *Let p be prime, let χ range over Dirichlet characters modulo p , and define*

$$S_a(p, \beta) = \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi) \quad (1 \leq a \leq p-1). \quad (5.14)$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p, \beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1), \quad \mu_{p,\beta}(0) = p^{-\beta}. \quad (5.15)$$

Consequently the prime-modulus functionals obtained by substituting (5.15) into $B_p(\beta)$ satisfy

$$\Sigma(\beta) = \lim_{\substack{p \rightarrow \infty \\ p \text{ prime}}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) = \sup_{p \text{ prime}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}). \quad (5.16)$$

Proof. For $a \not\equiv 0 \pmod{p}$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} \chi(n) = \begin{cases} 1, & n \equiv a \pmod{p}, \\ 0, & n \not\equiv a \pmod{p}. \end{cases} \quad (5.17)$$

Therefore

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{p}}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi). \quad (5.18)$$

Dividing by $\zeta(\beta)$ gives the nonzero residue formula, while

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}. \quad (5.19)$$

The functional identity follows by substituting these residue formulas into $B_p(\beta)$ and applying Theorem 5.2 along the prime sequence. $\square$

6 Euler-score identities

Theorem 6.1 (Euler-score decomposition). *For every $\beta > 1$,*

$$-(\log \zeta)'(\beta) = \mathbb{E}_{\beta}[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^{\beta}}. \quad (6.1)$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^{\beta}}. \quad (6.2)$$

Proof. The von Mangoldt identity is

$$\log n = \sum_{d|n} \Lambda(d). \quad (6.3)$$

By nonnegative summation,

$$\mathbb{E}_{\beta}[\log N] = \sum_{n \geq 1} \rho_{\beta}(n) \sum_{d|n} \Lambda(d) = \sum_{d \geq 1} \Lambda(d) \mathbb{P}_{\beta}(d | N). \quad (6.4)$$

The divisibility probability is

$$\mathbb{P}_{\beta}(d | N) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} (dm)^{-\beta} = d^{-\beta}. \quad (6.5)$$

Substitution into (6.4) proves (6.1); (6.2) follows from (3.5). $\square$

Proposition 6.2 (Finite Euler-score identity). *For every $A \subseteq \{1, \dots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md). \quad (6.6)$$

Proof. Reverse the order of summation:

$$\sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md) = \sum_{n \in A} \sum_{d|n} \Lambda(d) = \sum_{n \in A} \log n. \quad (6.7)$$

$\square$

7 Curvature and Mellin-Planck tools

Lemma 7.1 (Uniform positive-axis curvature bound). *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \text{Var}_s(\log N) < \frac{1}{3}. \quad (7.1)$$

Proof. By Proposition 3.3,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)} \right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s). \quad (7.2)$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}. \quad (7.3)$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$, so

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3} dx. \quad (7.4)$$

Since

$$\int_2^{\infty} \frac{(\log x)^2}{x^3} dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}, \quad (7.5)$$

we obtain

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}. \quad (7.6)$$

□

Definition 7.2 (Mellin-Planck partition function). For $s > 1$, define

$$M(s) = \Gamma(s)\zeta(s) = \int_0^{\infty} \frac{t^{s-1}}{e^t - 1} dt. \quad (7.7)$$

The corresponding probability law is

$$\nu_s(dt) = \frac{t^{s-1}}{(e^t - 1)M(s)} dt. \quad (7.8)$$

Proposition 7.3 (Mellin-Planck log-convexity). *For $s > 1$,*

$$\frac{d^2}{ds^2} \log M(s) = \text{Var}_{\nu_s}(\log t) \geq 0. \quad (7.9)$$

Proof. Differentiate the logarithm of the integral in (7.7) under the integral sign. The first derivative is the ν_s -expectation of $\log t$, and the second derivative is its variance:

$$\frac{d^2}{ds^2} \log M(s) = \mathbb{E}_{\nu_s}[(\log t)^2] - \mathbb{E}_{\nu_s}[\log t]^2. \quad (7.10)$$

□

Bulboacă and Zayed proved several monotonicity criteria for functions of the form $u(s)/(\Gamma(s + \rho)\Gamma(s))$ and asked for best possible positive values of ρ in their examples [5]. The next application gives the sharp endpoint in the base reciprocal Gamma-product case and the corresponding endpoint criterion for monotone logarithmic derivatives.

Theorem 7.4 (APP-0009: Sharp reciprocal Gamma-product monotonicity threshold). *Let $\psi = \Gamma'/\Gamma$, let γ be Euler's constant, and let $\rho_* > 0$ be the unique solution of*

$$\psi(1 + \rho_*) = \gamma. \quad (7.11)$$

For $\rho > -1$, define

$$\varphi_\rho(s) = \frac{1}{\Gamma(s + \rho)\Gamma(s)}, \quad s \geq 1. \quad (7.12)$$

Then φ_ρ is strictly decreasing on $[1, \infty)$ if and only if

$$\rho \geq \rho_*. \quad (7.13)$$

Equivalently,

$$W_\rho(s) = \Gamma(s + \rho)\Gamma(s) \quad (7.14)$$

is strictly increasing on $[1, \infty)$ if and only if (7.13) holds. Numerically,

$$\rho_* = 1.2583969670859318174106234224981693941 \dots \quad (7.15)$$

More generally, let $u : [1, \infty) \rightarrow (0, \infty)$ be differentiable and suppose

$$J(s) = \frac{u'(s)}{u(s)} \quad (7.16)$$

is nonincreasing on $[1, \infty)$. Then

$$\Phi_{\rho, u}(s) = \frac{u(s)}{\Gamma(s + \rho)\Gamma(s)} \quad (7.17)$$

is strictly decreasing on $[1, \infty)$ if and only if

$$\psi(1 + \rho) \geq \gamma + J(1). \quad (7.18)$$

Proof. The digamma function is strictly increasing on $(0, \infty)$, since

$$\psi'(x) = \sum_{k=0}^{\infty} \frac{1}{(x + k)^2} > 0. \quad (7.19)$$

Hence (7.11) has a unique positive solution.

For $s \geq 1$, the logarithmic derivative of (7.14) is

$$\frac{d}{ds} \log W_\rho(s) = \psi(s + \rho) + \psi(s). \quad (7.20)$$

The right-hand side is strictly increasing in s . Its minimum on $[1, \infty)$ is therefore

$$\psi(1 + \rho) + \psi(1) = \psi(1 + \rho) - \gamma. \quad (7.21)$$

If $\rho \geq \rho_*$, then (7.21) is nonnegative, and it is positive for every $s > 1$. Thus, for $1 \leq a < b$,

$$\log W_\rho(b) - \log W_\rho(a) = \int_a^b (\psi(s + \rho) + \psi(s)) ds > 0. \quad (7.22)$$

So W_ρ is strictly increasing and $\varphi_\rho = 1/W_\rho$ is strictly decreasing. Conversely, if $\rho < \rho_*$, then (7.21) is negative, so by continuity W_ρ decreases initially and φ_ρ increases initially. This proves the base equivalence.

For the general statement,

$$\frac{d}{ds} \log \Phi_{\rho,u}(s) = J(s) - \psi(s + \rho) - \psi(s). \quad (7.23)$$

The term J is nonincreasing by hypothesis, while $\psi(s + \rho) + \psi(s)$ is strictly increasing. Therefore the logarithmic derivative in (7.23) is strictly decreasing, and its maximum on $[1, \infty)$ is

$$J(1) - \psi(1 + \rho) - \psi(1) = J(1) - \psi(1 + \rho) + \gamma. \quad (7.24)$$

The function $\Phi_{\rho,u}$ is strictly decreasing exactly when this maximum is nonpositive, which is (7.18). If the inequality fails, the logarithmic derivative is positive near $s = 1$, so monotonicity fails. $\square$

Qi, Lim, and Nantomah asked whether the second derivative of $\log \Gamma(x) + \log \Gamma(1/x)$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers this in the affirmative by a reciprocal-Weierstrass Laplace kernel.

Theorem 7.5 (APP-0004: Complete monotonicity of reciprocal-Gamma curvature). *Let*

$$H(x) = \log \Gamma(x) + \log \Gamma(1/x), \quad x > 0. \quad (7.25)$$

Then H'' is strictly completely monotonic on $(0, \infty)$; that is,

$$(-1)^m \frac{d^m}{dx^m} H''(x) > 0 \quad (m \geq 0, x > 0). \quad (7.26)$$

Proof. The Weierstrass product gives, for $x > 0$,

$$\log \Gamma(x) = -\log x - \gamma x - \sum_{n=1}^{\infty} \left(\log \left(1 + \frac{x}{n} \right) - \frac{x}{n} \right). \quad (7.27)$$

After substituting $1/x$ in (7.27), adding the two formulas, and differentiating twice on compact subintervals of $(0, \infty)$, the logarithmic second-derivative terms cancel and

$$\begin{aligned} H''(x) &= \sum_{n=1}^{\infty} \frac{1}{(x+n)^2} - \frac{2\gamma}{x^3} \\ &\quad + \sum_{n=1}^{\infty} \left[\frac{1}{(x+1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} \right]. \end{aligned} \quad (7.28)$$

The elementary Laplace identities

$$\frac{1}{(x+a)^2} = \int_0^{\infty} t e^{-(x+a)t} dt, \quad \frac{1}{x^2} = \int_0^{\infty} t e^{-xt} dt, \quad \frac{2a}{x^3} = a \int_0^{\infty} t^2 e^{-xt} dt \quad (7.29)$$

show that, with $a = 1/n$,

$$\frac{1}{(x + 1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} = \int_0^\infty e^{-xt} t \left(e^{-t/n} - 1 + \frac{t}{n} \right) dt. \quad (7.30)$$

Also,

$$\sum_{n=1}^\infty \frac{1}{(x+n)^2} = \int_0^\infty e^{-xt} \frac{t}{e^t - 1} dt, \quad (7.31)$$

and

$$-\frac{2\gamma}{x^3} = \int_0^\infty e^{-xt} (-\gamma t^2) dt. \quad (7.32)$$

Combining (7.28)–(7.32) gives

$$H''(x) = \int_0^\infty e^{-xt} K(t) dt, \quad (7.33)$$

where

$$K(t) = \frac{t}{e^t - 1} + t \sum_{n=1}^\infty \left(e^{-t/n} - 1 + \frac{t}{n} \right) - \gamma t^2. \quad (7.34)$$

It remains to prove $K(t) > 0$. Put

$$\phi(u) = e^{-u} - 1 + u. \quad (7.35)$$

Then $\phi \geq 0$, $\phi' = 1 - e^{-u} \geq 0$, and $u \mapsto \phi(t/u)$ is positive and decreasing on $[1, \infty)$. Therefore

$$\sum_{n=1}^\infty \phi(t/n) \geq \int_1^\infty \phi(t/u) du. \quad (7.36)$$

With $v = t/u$, integration by parts gives

$$\int_1^\infty \phi(t/u) du = t \int_0^t \frac{\phi(v)}{v^2} dv = -\phi(t) + t \int_0^t \frac{1 - e^{-v}}{v} dv. \quad (7.37)$$

Using

$$\int_0^t \frac{1 - e^{-v}}{v} dv = \gamma + \log t + E_1(t), \quad E_1(t) = \int_t^\infty \frac{e^{-v}}{v} dv, \quad (7.38)$$

we obtain

$$\frac{K(t)}{t} \geq \frac{1}{e^t - 1} - \phi(t) + t \log t + t E_1(t). \quad (7.39)$$

Since $E_1(t) > 0$ and $\log t \geq 1 - 1/t$ for $t > 0$,

$$\frac{K(t)}{t} > \frac{1}{e^t - 1} - \phi(t) + t - 1 = \frac{1}{e^t - 1} - e^{-t} = \frac{e^{-t}}{e^t - 1} > 0. \quad (7.40)$$

Thus $K(t) > 0$ for $t > 0$. Differentiating under the Laplace integral in (7.33) gives

$$(-1)^m \frac{d^m}{dx^m} H''(x) = \int_0^\infty t^m e^{-xt} K(t) dt > 0. \quad (7.41)$$

This proves (7.26). □

8 External zeta inequality proofs

Theorem 8.1 (APP-0002: Affirmative solution of Nantomah zeta positivity problem). *For every $n \in \mathbb{N}$,*

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \quad (8.1)$$

Proof. Put $s = n+1 \geq 2$ and $Z_s = \zeta(s)$. The expression in (8.1) becomes

$$K_s = (s+1)Z_s Z_{s+2} - sZ_{s+1}^2 - Z_s Z_{s+1}. \quad (8.2)$$

Let $X(m) = 1/m$ under ρ_s . By (3.51),

$$\mathbb{E}_s[X] = \frac{Z_{s+1}}{Z_s}, \quad \mathbb{E}_s[X^2] = \frac{Z_{s+2}}{Z_s}. \quad (8.3)$$

Thus

$$\frac{K_s}{Z_s^2} = (s+1)\mathbb{E}_s[X^2] - s\mathbb{E}_s[X]^2 - \mathbb{E}_s[X] = (s+1)\text{Var}_s(X) + \mathbb{E}_s[X]^2 - \mathbb{E}_s[X]. \quad (8.4)$$

Write $a = \zeta(s) - 1$, $b = \zeta(s+1) - 1$, and $c = \zeta(s+2) - 1$. Expanding gives

$$K_s = L_s + Q_s, \quad L_s = sa + (s+1)c - (2s+1)b, \quad Q_s = (s+1)ac - sb^2 - ab. \quad (8.5)$$

The linear part is termwise positive:

$$L_s = \sum_{m=2}^{\infty} m^{-s} \frac{(m-1)(s(m-1)-1)}{m^2} \geq 2^{-s} \frac{s-1}{4}. \quad (8.6)$$

Since $b \leq a/2$ and $c \geq 0$,

$$Q_s \geq -sb^2 - ab \geq -\frac{s+2}{4}a^2. \quad (8.7)$$

For $s \geq 4$,

$$a = \sum_{m=2}^{\infty} m^{-s} \leq 2^{-s} + \int_2^{\infty} x^{-s} dx = 2^{-s} \left(1 + \frac{2}{s-1}\right). \quad (8.8)$$

Equations (8.6), (8.7), and (8.8) imply

$$K_s \geq \frac{1}{4} \left[2^{-s}(s-1) - (s+2)2^{-2s} \left(1 + \frac{2}{s-1}\right)^2 \right]. \quad (8.9)$$

It is therefore enough to show

$$2^s(s-1) > (s+2) \left(1 + \frac{2}{s-1}\right)^2. \quad (8.10)$$

Equivalently,

$$F(s) = \frac{2^s(s-1)^3}{(s+2)(s+1)^2} > 1. \quad (8.11)$$

Now $F(4) = 72/25 > 1$, and

$$\frac{d}{ds} \log F(s) = \log 2 + \frac{3}{s-1} - \frac{1}{s+2} - \frac{2}{s+1} > 0 \quad (s \geq 4). \quad (8.12)$$

Thus $K_s > 0$ for $s \geq 4$.

For $s = 2$,

$$K_2 = 3\zeta(2)\zeta(4) - 2\zeta(3)^2 - \zeta(2)\zeta(3). \quad (8.13)$$

Using $\zeta(3) < 5/4$, $\zeta(2) = \pi^2/6$, and $\zeta(4) = \pi^4/90$,

$$K_2 > \frac{\pi^6}{180} - \frac{5\pi^2}{24} - \frac{25}{8} > \frac{10415360504209}{18000000000000} > 0. \quad (8.14)$$

For $s = 3$, the estimates $\zeta(3) > 251/216$, $\zeta(5) > 8051/7776$, and $\zeta(4) < 13/12$ yield

$$K_3 > \frac{251}{216} \left(4 \cdot \frac{8051}{7776} - \frac{13}{12} \right) - 3 \left(\frac{13}{12} \right)^2 = \frac{13783}{419904} > 0. \quad (8.15)$$

Hence $K_s > 0$ for all integers $s \geq 2$, proving the theorem. This answers Nantomah's question [1]. $\square$

Theorem 8.2 (APP-0001: Alzer-Kwong convexity and concavity pattern for reciprocal zeta). *Let $F(x) = 1/\zeta(x)$. For every integer $n \geq 1$,*

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \quad (8.16)$$

and

$$F''(x) < 0 \quad \text{on } (-4n-2, -4n). \quad (8.17)$$

Proof. Put $x = -u$, where $u > 2$. The functional equation gives

$$\zeta(-u) = -2^{-u}\pi^{-u-1} \sin\left(\frac{\pi u}{2}\right) \Gamma(u+1)\zeta(u+1), \quad (8.18)$$

and hence

$$\frac{1}{\zeta(-u)} = -\frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)\sin(\pi u/2)}. \quad (8.19)$$

Define the positive function

$$G(u) = \frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)|\sin(\pi u/2)|}. \quad (8.20)$$

On $u \in (4n-2, 4n)$, $\sin(\pi u/2) < 0$, so $F(-u) = G(u)$. On $u \in (4n, 4n+2)$, $\sin(\pi u/2) > 0$, so $F(-u) = -G(u)$. Since second derivatives are unchanged by $x = -u$, it remains to prove $G''(u) > 0$.

Let $\ell(u) = \log G(u)$. Then

$$\ell(u) = u \log 2 + (u+1) \log \pi - \log \Gamma(u+1) - \log \zeta(u+1) - \log \left| \sin\left(\frac{\pi u}{2}\right) \right|. \quad (8.21)$$

Differentiating twice gives

$$\ell''(u) = -\psi'(u+1) - (\log \zeta)''(u+1) + \frac{\pi^2}{4} \csc^2\left(\frac{\pi u}{2}\right). \quad (8.22)$$

With $s = u+1 > 3$,

$$\psi'(s) = \sum_{k=0}^{\infty} \frac{1}{(s+k)^2} < \int_{s-1}^{\infty} \frac{dt}{t^2} = \frac{1}{s-1} \leq \frac{1}{2}. \quad (8.23)$$

Combining (7.1), (8.23), and $\csc^2(\pi u/2) \geq 1$, we get

$$\ell''(u) > \frac{\pi^2}{4} - \frac{1}{2} - \frac{1}{3} = \frac{\pi^2}{4} - \frac{5}{6} > 0. \quad (8.24)$$

Since $G > 0$,

$$G''(u) = G(u)(\ell''(u) + \ell'(u)^2) > 0. \quad (8.25)$$

The two signs in (8.16) and (8.17) now follow from $F(-u) = G(u)$ on $u \in (4n-2, 4n)$ and $F(-u) = -G(u)$ on $u \in (4n, 4n+2)$. This proves the Alzer–Kwong sign pattern [3]. $\square$

Theorem 8.3 (APP-0003: Generalized Holder inequality for Gamma zeta). *Let $m \geq 2$, $r \geq 1$, $x_1, \dots, x_m > 1$, and $p_1, \dots, p_m > 1$ satisfy*

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}. \quad (8.26)$$

Define

$$T = r \sum_{i=1}^m \frac{x_i}{p_i}. \quad (8.27)$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}}{\Gamma(T)}. \quad (8.28)$$

Proof. Set

$$\lambda_i = \frac{r}{p_i}. \quad (8.29)$$

Then $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and $T = \sum_i \lambda_i x_i > 1$. Define

$$q_i = \frac{p_i}{r}. \quad (8.30)$$

Then $\sum_i 1/q_i = 1$, and the assumptions imply $q_i > 1$ for all i ; otherwise one term $1/p_i$ would already be at least $1/r$, leaving no room for the remaining positive terms in (8.26).

For $t > 0$, let

$$f_i(t) = \left(\frac{t^{x_i-1}}{e^t - 1} \right)^{1/p_i}. \quad (8.31)$$

Then

$$\left(\prod_{i=1}^m f_i(t) \right)^r = \frac{t^{r \sum_i (x_i-1)/p_i}}{e^t - 1} = \frac{t^{T-1}}{e^t - 1}. \quad (8.32)$$

Consequently,

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r(0, \infty)}^r = \int_0^\infty \frac{t^{T-1}}{e^t - 1} dt = \Gamma(T) \zeta(T). \quad (8.33)$$

Generalized Holder gives

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r} \leq \prod_{i=1}^m \|f_i\|_{L^{p_i}}. \quad (8.34)$$

Moreover,

$$\|f_i\|_{L^{p_i}}^{p_i} = \int_0^\infty \frac{t^{x_i-1}}{e^t - 1} dt = \Gamma(x_i) \zeta(x_i). \quad (8.35)$$

Combining (8.33), (8.34), and (8.35), then raising to the power r , gives

$$\Gamma(T)\zeta(T) \leq \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}. \quad (8.36)$$

Division by $\Gamma(T)$ proves (8.28). This solves the generalized Holder problem in Sroysang's notation, where his $\xi(s)$ is $\zeta(s)$ for $s > 1$ [2]. $\square$

Remark 8.4 (Free-energy interpretation of the Sroysang inequality). The same result follows from log-convexity of $M(s) = \Gamma(s)\zeta(s)$. With $\lambda_i = r/p_i$ and $T = \sum_i \lambda_i x_i$, Jensen's inequality for $\log M$ gives

$$M(T) \leq \prod_{i=1}^m M(x_i)^{\lambda_i} = \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}, \quad (8.37)$$

which is equivalent to (8.28).

9 Conclusion

The zeta-law normalization gives a single language for several identities and inequalities involving ζ . At the structural level, the paper proves the successor entropy formula (5.4) and its modular resolution (5.5), the Euler-score decomposition (6.1), the Mellin-Planck variance identity (7.9), the reciprocal Gamma-product threshold theorem (7.4), the reciprocal-Gamma complete-monotonicity theorem (7.5), the reciprocal-digamma product theorem (3.4), the higher-polygamma complete-monotonicity counterexample (3.5), and the tail partition formulas (4.5) and (4.23). These results are not only formal translations: they supply direct estimates or obstructions for the nine labelled applications in Section 2. APP-0001 follows from the functional equation and a positive-axis curvature estimate; APP-0002 follows from a moment split and tail bounds; APP-0003 follows from the Planck-kernel integral representation; APP-0004 follows from a positive reciprocal-Gamma Laplace kernel; APP-0005 follows from a rational telescoping enclosure; APP-0006 follows from adjacent corrected asymptotic truncations; APP-0007 follows from a grouped reciprocal-digamma Laplace kernel; APP-0008 follows from a dominant-summand argument proving $P_n'''(2) > 0$ for all $n \geq 29$, together with a rational interval certificate showing $P_7^{(6)}(3) < 0$; and APP-0009 follows from an endpoint logarithmic-derivative certificate for reciprocal Gamma products. The remaining useful directions are to settle the weaker convexity question for higher polygamma products, sharpen the polynomial Gamma-product threshold problem beyond monotone u'/u , and sharpen the modular entropy side, especially the prime-modulus functionals in (5.16), where additive successor entropy is encoded by finite Dirichlet L -value data.

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